Gen_X：可行技术路线与运行记录

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Gen_X 技术路线：学习构型概率分布，再生成结构系综

调研截止：2026-10-22。本文给出整体科学路线。最初完成 Bash/Slurm CPU 冒烟测试；随后已实际运行 bgmat 预训练生成器，得到 8 个 216 粒子的 mW 冰构型，详见真实生成结果。这推进了模型接入，但训练、DFT 标注和收敛的科学分布验证仍未执行。初始调度证据由 smoke_results.md 保留。

1. 研究对象与起步选择

核心任务是学习指定条件下的构型分布，再独立采样多个结构。令构型为$X = (\mathbf{R}, \mathbf{H}, \mathbf{Z}, \mathcal{G})$，分别表示坐标、盒矩阵、元素及可选分子拓扑；条件$c$ 包括组分、温度、密度、相态或制备历史。生成器学习$q_\theta(X \mid c)$，输出结构系综及来源记录。单个低能结构、几何合法的坐标文件、与目标相似的 RDF，都不足以证明学到了完整分布。

固定粒子数和盒子的经典平衡体系可用$p_\phi(\mathbf{R} \mid c) \propto \exp[-\beta U_\phi(\mathbf{R})]$ 定义目标，其中$\beta = (k_B T)^{-1}$。这是给定势下的分布，不自动等于真实材料。NPT 还需处理体积和坐标测度。非晶玻璃通常有制备历史依赖，应学习$p(X \mid$ 组分、密度、冷却率、老化历史$)$，不能未经验证就称为平衡 Boltzmann 分布。

路线与顺序	近期任务	选择理由与明确边界
A：DM2 非晶，优先	用上游 SiO_2 权重跑通生成和结构统计；随后检验冷却条件及尺寸迁移	上游提供权重、数据与示例，启动成本较低；不代表已在本机运行，也不代表任意非晶通用
B：scalable BG / bgmat，补充	复现单珠 mW 冰的采样、权重与有效样本量，检验小到大迁移	适合回答“采样是否服从目标分布”；mW 一粒子代表一水分子，冰的结果不能外推为全原子液态水
C：全原子水，后续主攻	先冻结已验证水势，再训练周期性条件生成器	可研究氢键网络及状态迁移，但需额外解决分子朝向、长程作用和势适用域
D：聚合物，替代方向	固定化学组成与 CG 映射，生成全原子细节并比较平衡成本	适合工程应用；需保持链连通、立构和缠结，不能由局部几何推断整体已平衡

DM2 已正式发表，在线日期为 2025-12-10，卷期归于 2026；其模型针对非晶硅氧网络和金属玻璃，条件生成涉及冷却率。论文与官方权重、代码支持路线 A。scalable BG 在 mW 立方冰上报告从 216 粒子训练向 512、1000、1728 粒子迁移；这些是文献结果，构成路线 B 的复现目标。论文；bgmat。

2. 生成模型家族如何选

“最新”不等于在当前数据和验证目标下最合适。以下比较聚焦采样目标、密度可计算性与结构约束；生成质量和速度必须在相同材料、相同参考数据上实测。

模型家族	学习目标与采样	本项目取舍及代表工作
扩散 / score 模型	预测扰动噪声或含噪分布的 score；多步去噪生成	适合复杂多峰结构与条件生成；推理需多次网络计算，普通去噪损失不保证精确 Boltzmann 权重。DM2 是首选落地实例
Flow matching	回归从基分布通往数据的速度场；积分 ODE 采样	训练形式简洁，适合周期流形及等变图；流的“时间”不是物理时间。Riemannian FM 已用于单珠冰，仍需检查 ODE、密度与重加权误差
离散可逆 normalizing flow	用可逆变换和 Jacobian 得到密度；最大似然或能量与熵目标训练	适合重要性重加权、自由能与 ESS 验证；表达能力、粒子排列及大体系权重退化是限制。scalable BG 属于此路线

模型家族	学习目标与采样	本项目取舍及代表工作
自回归	分解为有序条件概率的乘积，逐点、逐片或逐链生成	易加入局部条件和拓扑，但生成顺序、边界闭合和误差累积需检验。MAP 已展示二维无序结构的多尺度生成；不能直接当作三维水的证据
VAE / GAN	VAE 优化重构与潜变量正则；GAN 用对抗目标匹配分布	可做压缩、回映射和快速基线；VAE 可能丢失细节，GAN 可能遗漏模式，通常没有可直接重加权的构型密度。GlassVAE、CryinGAN、DeepBackmap 提供实际比较对象

近期可核实实例包括：Hoffmann 等的 Riemannian flow matching（AI4MAT/ICLR 2026 workshop）；Chen 等的 GlassVAE（AAAI 2026）；以及 Dismai-Bench / CryinGAN（2024）。后者说明部分 GAN 能超过某些坐标扩散模型，模型名称不能替代同任务比较。MAP 的二维工作（2020）和 DeepBackmap（2020）作为有明确适用范围的已有基线。

Flow matching 是训练方法，连续 normalizing flow 是模型形式，两者并非互斥类别。2026-10 更新的非晶 Boltzmann generator 工作发现，连续流的似然数值积分可能破坏可逆性和热力学重加权。因此不能仅凭连续模型在数学上可逆就声称数值结果无偏。Grenioux 等，v4。

机器学习势与潜空间也应分开：势通常学习$X \rightarrow U_\phi, \mathbf{F}_\phi$；其内部特征不是可逆坐标编码。生成器学习$z, c \rightarrow X$；只有采用编码器与解码器时，才涉及压缩潜空间。VAE 压缩、CG 物理映射、normalizing flow 的可逆基变量是不同概念。局域势可处理更多原子，并不保证生成器能恢复大盒子的长波涨落。

3. 数据与结构约定

每个样本必须保存坐标、元素、盒子、PBC 标志、单位、状态/制备条件、来源轨迹与种子、原始文件哈希，以及势、生成器和处理程序版本。能量、力、应力应区分真实参考标签和模型预测。另存原始生成、筛选后、松弛后结构及拒绝原因，避免后处理掩盖生成缺陷。最初优先固定组分、粒子数和盒子，再研究可变体积或尺寸。

风险	数据和表示要求	验证方式
轨迹与结构泄漏	按独立轨迹、母构型、冷却重复或聚合物母链分组划分；相关邻帧和增强副本归同组	检查跨集合近邻重复；专设留出状态、历史和尺寸，不用随机拆帧代替迁移验证
周期边界	分数坐标与盒矩阵对应，邻居边带周期位移；噪声/插值尊重环面几何	整体平移、跨边界移动及一致旋转盒子后的结果应相容；检查盒变形与最短距离
水与聚合物拓扑	水保持分子身份，明确刚性或柔性表示；聚合物保存键、立构、链长和 CG 对应	键长/键角、断链、跨边界整链重建；水朝向与氢键，聚合物回转半径及缠结诊断
非晶历史	显式记录淬火曲线、冷却率、老化、密度和最终温度	按历史比较环、配位、自由体积与力学响应；把未见冷却率标为外推

标准化参数只从训练集估计。验证集用于选模型与阈值；冻结测试集用于最终报告。上游模型已见过的数据只能支持复现检查，不能充当新的泛化证据。迁移实验应区分“同一势的新构型”“新状态”“新体系尺寸”和“新化学组成”。

4. 阶段、输出与停止条件

各阶段按前置证据推进；A、B 是互补验证，不必先把所有模型都训练一遍。当前因 CPU 可用性先完成了 B 路线的 8 构型推理示例；下表关于系综、自由能和尺寸迁移的完整科学验收仍属于计划。

阶段	输入与产物	验收及停止条件
S0：调度与溯源	版本化路线配置、明确标为 synthetic 的小样例；输出唯一 run ID、manifest、日志及候选队列	本地错误传播、禁止覆盖、Slurm 退出码与产物一致；dry-run 不算提交成功。失败则停在软件修复
S1：DM2 最小推理	固定提交、权重哈希、上游示例；输出原始构型、采样参数和性能测量	先与上游同体系同设置比较；几何异常、依赖不兼容或无法复现时不扩大规模
S2：DM2 系综验证	多独立种子、参考结构及留出条件；输出配位/角/环分布、RDF、多样性与松弛变化	与独立参考区间比较；必须报告拒绝率。若仅 RDF 接近而网络或力学失真，先改生成器/数据
S3：BG 概率与尺寸	mW 冰小体系参考与受控大体系；输出密度/权重、ESS、结构和能量分布	重加权量在增样本与数值精度检查下收敛；ESS 退化或似然误差主导时，暂停大体系/自由能结论
S4：全原子水或聚合物	水专用势/固定化学力场、独立参考轨迹、分子约束；输出条件生成器与采样系综	与同势 MD 和简单初始化/回映射比较质量及总成本；不稳定、拓扑破坏或模式遗漏则停在该域
S5：反馈迭代与迁移	经审查的缺口候选、独立新数据；输出新数据集及模型版本	冻结测试不回流；新模型同时改善目标缺口且无明显回退后再晋级，否则保留原版本

进入科学运行前，在配置中冻结状态范围、主要观测量、误差容限及参考采样长度。容限依据独立参考重复的统计误差与实际科学用途设定，不能在看到生成结果后修改。保留样本须满足硬约束；总体成功率以全部尝试为分母。样本量逐步扩大到主要观测的置信区间达到预设宽度，不把任意固定样本数当作充分性保证。

5. 两条独立的反馈回环

生成器回环：找分布缺口，再补可信样本。生成构型后进行几何筛选，按局部环境新颖性、状态覆盖和高阶统计偏差选候选；从相应条件启动独立参考采样或补充已有可信数据，加入训练集后更新生成器。只把自身生成样本反复加入训练，容易放大遗漏和偏差，不能称为获取了新的真实信息。松弛结果也不自动是平衡分布样本。

势的主动学习回环：仅在势可能失效时触发。第一版冻结已验证势。发现域外局部环境、经校准的势委员会分歧、异常力或持续参考偏差时，先去重并聚类选取代表结构，再用独立 DFT/目标电子结构方法计算能量、力及需要的应力。只有参考计算确认误差或覆盖缺口，并满足预定预算与数据质量条件，才扩充势训练集并微调；随后用冻结测试与物性重新验收。委员会分歧是选点信号，不是真实误差标签，单势能量低也不是可信证明。

势更新后重新生成或校正参考系综，并标记旧样本对应的势版本；不能把不同目标势的分布无条件混为同一训练集。对真实水还需区分电子结构近似、核量子效应和经典采样误差，避免把所有实验偏差都归因于神经网络拟合。

两条回环共享候选的来源、状态与版本，但拥有独立的触发条件和晋级门槛。本轮 `synthetic fixtures → QA → candidates.jsonl` 只验证文件与路由；其分数不是已校准不确定度，也不会自动提交 DFT 或重训。独立标签、独立参考系综和留出实验用于避免“同一个模型生成、打分并证明自己正确”。

6. 科学验收、资源与可发表结果

验证按四层报告：几何和拓扑合法性；与参考系综的分布一致性；跨状态/历史/尺寸迁移；独立物理观测。非晶看网络和力学，水看分子朝向与氢键，聚合物看链尺度及缠结。低能筛选后的漂亮结构图、短 MD 稳定性或 RDF 单项通过都不能代替这四层证据。

对密度可计算且权重可靠的模型，可报告$\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$，并检查权重尾部与重复间波动；自归一化估计的有限样本偏差仍需控制。普通扩散/GAN 若无可靠密度，不能编造同类重要性 ESS，宜用独立重复、覆盖和观测误差评估。静态生成样本没有物理时间顺序；扩散系数和寿命需另跑动力学，后续成本计入性能比较。

实验约束可以作为后续研究：对系综平均$\langle S(q)\rangle$ 而非每帧曲线施加含误差的约束，并留出另一实验通道。由同一$S(q)$ 变换得到的$g(r)$ 不算独立验证；最终要检验多种物理先验下哪些结构结论仍稳定。单凭拟合好不能声称唯一反演水网络。已有 RMC 反例说明这一边界。

登录环境用于编辑与轻量检查，训练、模拟和推理统一经 Slurm。最初完成了短 CPU smoke，随后用隔离的 bgmat 环境和 2 CPU 跑通真实生成；GPU 仍未申请或使用。已有 `deepmd3.2` 环境不代表能运行所有生成器，应逐路线核对依赖和模型格式。当前 CPU-only 示例利用 GPU 分区的空闲 CPU，具体分区与资源政策以集群实际配置为准。

算力预算通过 S1 小任务测量峰值显存、每步/每构型耗时、后处理与参考采样成本，再对尺寸和批量作实测扩展；不预填未经验证的 GPU 小时。报告总成本应包含数据生成、训练、失败样本、松弛和验证，并与达到同一统计误差的 MD/传统回映射比较。

近期可交付结果是可复现的生成与验收流程；研究贡献则应来自可量化的新能力，例如条件非晶的网络与力学同时可信、大体系权重退化得到控制，或全原子水的高阶分布在留出状态下稳定。若收益仅为重新生成训练集附近结构，或成本优势在计入后处理后消失，就停止扩大叙事，保留有证据的能力边界。

CPU smoke execution record

Executed on 2026-10-22 in `/home/qiuyong/bza/H2O/Gen_X`. Scope: Bash orchestration, configuration checks, synthetic geometry QA and candidate queue, an initial Slurm CPU job and a second CPU job verifying review fixes. No external generator, potential inference, GPU computation, training, model/data download, or dependency installation was performed. Historical runs are preserved; the reviewed-version results are identified separately below.

Entry points

```
# Lightweight software smoke; a unique run ID is generated when omitted.
bash scripts/local_smoke.sh --config configs/dm2_smoke.json
bash scripts/local_smoke.sh --config configs/bgmat_smoke.json
bash scripts/local_smoke.sh --config configs/mbpol_water_smoke.json

# Validate inputs and print the argv array without creating a run or submitting.
bash scripts/submit_smoke.sh --config configs/dm2_smoke.json --partition debug --dry-run

# Submit a new, CPU-only smoke. This creates a real Slurm job.
bash scripts/submit_smoke.sh --config configs/dm2_smoke.json --partition debug

# Tests use only Python's standard library; no scheduler submission occurs.
PYTHONDONTWRITEBYTECODE=1 /usr/bin/python3 -m unittest discover -s tests -v
```

The Bash launchers default to `/usr/bin/python3`; `GENX_PYTHON` can select an absolute compatible interpreter. The submitter carries that absolute interpreter path into the batch job. Python 3.6 or newer is required by the implementation; the executed login and compute runtimes were Python 3.12.15 and 3.12.13, respectively. No claim of execution on Python 3.6 is made.

Use `--output-root PATH` inside this repository and optional `--run-id NAME` to choose output names. Existing run directories are rejected, even if empty or previously failed. Paths containing spaces are passed as individual arguments. Configuration fields are not evaluated as shell code. Each run snapshots its input configuration and fixture, records hashes and the exact command array, and creates artifacts with exclusive file creation. The reviewed implementation derives the original and resolved snapshots from the same validated input bytes and verifies the exact resolved-config hash before parsing those bytes for execution. The worker also checks resolved-path containment before reading or writing run files, including when invoked directly. Worker replay is refused. Slurm writes its own scheduler stdout/stderr into the newly reserved run directory.

The Slurm request uses the generic `debug` partition, one node, one task, one CPU, 128 MB and a two-minute time limit. It does not bind to a specific node or request GPU resources. Inherited `SBATCH_*` variables are removed before invoking `sbatch`, and the batch job receives `--export=NONE`; the batch shell also clears `PYTHONHOME`, `PYTHONPATH` and `LD_LIBRARY_PATH`. The reviewed version explicitly requests `--open-mode=append` and `--no-requeue` to preserve scheduler logs and avoid automatic replay of a worker that refuses replay. All output goes directly to the shared project filesystem; node-local scratch is not used.

Tests and local execution

The initial ten behavior tests cover unknown route, missing required metadata, paths with spaces and periodic overlap detection, output collision, worker subprocess failure, local dry-run, small CPU Slurm dry-run, scheduler rejection, run-ID path traversal, and rejection of a training mode. The scheduler-rejection test uses a deliberately failing local `sbatch` substitute and checks exit status 17 and a failure manifest; it does not contact the cluster.

Initial test run before launcher implementation: 10 tests, 8 failures and 1 error (missing launchers/manifest); one negative-only traversal test already passed. The saved initial red log makes that distinction explicit. After initial implementation: all 10 passed in the green log. Review added four regression tests: editing the original config between validation and staging, modifying the resolved config before execution, Slurm append/no-requeue flags, and direct worker access via a symlink resolving outside the workspace. All four failed before the fixes (regression red); the full

revised suite passes **14/14** (regression green). Tests create temporary directories inside `tests/` and remove only those task-owned test directories on completion.

Executed local commands:

```
bash scripts/local_smoke.sh --config configs/dm2_smoke.json --run-id local_dm2_20261022
bash scripts/local_smoke.sh --config configs/bgmat_smoke.json --run-id local_bgmat_20261022
bash scripts/local_smoke.sh --config configs/mbpol_water_smoke.json --run-id local_mbpol_20261022
bash scripts/submit_smoke.sh --config configs/dm2_smoke.json --run-id dryrun_dm2_20261022 --dry-run
bash -n scripts/local_smoke.sh scripts/submit_smoke.sh scripts/slurm_smoke.sh
```

All three local runs returned zero and produced three synthetic QA records, one illustrative QA pass and two flagged candidates. Evidence:

Route	Manifest	Meaning
DM2	local DM2 manifest	Hand-authored silicon coordinates; no DM2 invocation
bgmat	local bgmat manifest	Hand-authored single-bead coordinates; no bgmat invocation
DNN@MB-pol water	local water manifest	Hand-authored O/H coordinates; no potential or generator invocation

The Slurm dry-run record stores the planned argument array. It did not create `runs/dryrun_dm2_20261022` or submit a job. Bash syntax checks returned zero.

Original Slurm execution

The initial implementation submitted one actual smoke job, before independent-review fixes:

```
bash scripts/submit_smoke.sh --config configs/dm2_smoke.json --run-id slurm_dm2_20261022 --partition debug
squeue -j 96379 -o '%.18i %.9P %.20j %.8T %.10M %.6D %R'
sacct -j 96379 --format=JobID,JobName,Partition,AllocCPUS,ReqMem,Elapsed,State,ExitCode,NodeList,ReqTRES,AllocTRES -P
scontrol show job 96379
```

Field	Observed result
Job ID	`96379`
Partition / allocated node	`debug` / `node04` (scheduler selected)
Allocation	1 CPU, 128 MB; no GPU TRES
Requested limit	`00:02:00`
Accounted elapsed time	`00:00:05`
Job / batch / extern state	`COMPLETED`
Job / batch / extern exit code	`0:0`
Worker finish time	`2026-10-22T07:50:38.519045+00:00` (15:50:38 Asia/Shanghai)
Worker result	3 synthetic samples; 1 QA pass; 2 flagged candidates

Evidence: submission record, accounting, scheduler details, stdout, stderr, terminal manifest, and candidate queue. Scheduler stderr was empty. The manifest records both Python versions and the worker host; accounting independently confirms successful completion. Existing jobs were inspected only and not modified.

This run remains a valid successful smoke of its recorded implementation. Its script hash differs from the reviewed current script. The exact original script snapshot retains SHA-256

f4ce17d74d62d8cd9cedde1ba72716615fa9fe95afb4fbdb6fcb8ac63cd5fff1, matching the original manifests. No earlier results or logs were modified during the review fixes.

Reviewed-version execution

After the three review fixes and the 14-test passing suite, all three route smokes were run again into new directories:

```
bash scripts/local_smoke.sh --config configs/dm2_smoke.json --run-id local_dm2_review_20261022
bash scripts/local_smoke.sh --config configs/bgmat_smoke.json --run-id local_bgmat_review_20261022
bash scripts/local_smoke.sh --config configs/mbpol_water_smoke.json --run-id local_mbpol_review_20261022
bash scripts/submit_smoke.sh --config configs/dm2_smoke.json --run-id dryrun_dm2_review_20261022 --dry-run
bash scripts/submit_smoke.sh --config configs/dm2_smoke.json --run-id slurm_dm2_review_20261022 --partition debug
sacct -j 96390 --format=JobID,JobName,Partition,AllocCPUS,ReqMem,Elapsed,State,ExitCode,NodeList,ReqTRES,AllocTRES -P
scontrol show job 96390
```

The revised DM2, bgmat, and water local manifests all report successful software smoke with three synthetic records, one illustrative QA pass, and two candidates. The revised dry-run command includes both `--open-mode=append` and `--no-requeue` and created no dry-run directory.

One additional actual CPU job verified the final script; the task therefore submitted **two actual CPU jobs in total**:

Field	Reviewed-version result
Job ID	96390
Partition / allocated node	debug / node04 (scheduler selected)
Allocation	1 CPU, 128 MB; no GPU TRES
Requested limit	00:02:00
Accounted elapsed time	00:00:04
Job / batch / extern state	COMPLETED
Job / batch / extern exit code	0:0
Requeue setting	Requeue=0
Worker finish time	2026-10-22T08:00:07.479414+00:00 (16:00:07 Asia/Shanghai)
Worker result	3 synthetic samples; 1 QA pass; 2 flagged candidates

The current script SHA-256 is 7fbcdbb654014889c910bc65f5d2388a7e9076405a66b5bba97a823164a89069, matching all four revised-version manifests. The Slurm manifest also records resolved-config SHA-256 31b9c844417a5904bc53cd120617eb0060349cfe22096a6a315e77592e48466c.

Evidence: submission record, command array, accounting, scheduler details, stdout, stderr, terminal manifest, and candidate queue. Scheduler stderr was empty. Automatic restart behavior was not deliberately triggered; the accepted scheduler request, Requeue=0, and regression test verify the configured prevention measures.

Interpretation and limits

completed describes the software smoke. Every manifest explicitly retains model_inference: not_run and scientific_validation: not_run. Route configurations provide upstream sources, model roles, unresolved entrypoints and prerequisites. They do not provide runnable adapters for the cited external models. DNN@MB-pol is registered as a potential teacher for a future all-atom generator; it is not represented as a generator.

The illustrative QA computes minimum pair distance under orthorhombic periodic boundaries and flags coordinates outside the primary cell. The common 0.7 Å threshold is a test parameter, not a validated element-specific or molecular acceptance criterion. Passing these checks does not establish physical validity. Fixtures and every QA/candidate record carry explicit synthetic

provenance; bgmat bead data are not called all-atom water. The queue priority is the count of QA flags, not uncertainty, model error, Boltzmann weight, or acquisition value. Candidate actions are `manual_review_only`; no DFT, labeling, retraining, simulation or external submission is triggered by the queue.

Pending scientific milestones: pin and review upstream code/weights/licenses; establish runnable inference environments and unit/species contracts; reproduce published examples; evaluate energy/forces and distribution/ensemble observables; design calibrated candidate selection and separate potential/generator feedback loops. Those stages need their own successful run evidence.